

Supplementary Material for MnL with Reactive Blocks

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1 Introduction

Block relation: - Parent - Child

Block state: - Close (for all input had connected with child block) -
Partial (there is exist input had not connected with child block)

2 Typing Rules

Typing

$$\boxed{\Gamma \vdash b : BT}$$

$$\boxed{\Gamma \vdash incomplete : ST \otimes *}$$

(BT-INCDYNAMIC)

$$\boxed{\Gamma \vdash incomplete : ST \otimes T}$$

(BT-INCSTATIC)

Syntax	Terms :	Syntax Metamorphosis
$t ::=$	Terms :	$\boxed{\text{term} \rightarrow \text{block}}$
$\quad \quad \quad \text{con}$	contant	$\text{con} ::= \text{Constants :}$
$\quad \quad \quad \text{pat}$	pattern	string
$\quad \quad \quad \text{dec}$	declaration	
$\quad \quad \quad \text{exp}$	expression	$\text{pat} ::= \text{Patterns :}$
$\quad \quad \quad \text{param}$	parameter	string
$\quad \quad \quad \text{uid}$	identifier	
$\quad \quad \quad \text{mtc}$	match	
$\quad \quad \quad \text{field}$	field	
$\text{con} ::=$	Constants :	
$\quad \text{string } v$	string	
$\quad \text{character } v$	character	
$\quad \text{boolean } v$	boolean	
$\quad \text{number } v$	number	
$\quad \text{unit } v$	unit	
$v ::=$	Values :	
$\quad x$	identifier	
$\quad \text{t} \rightarrow \text{t}$	function	
$\quad ()$	unit	
$\quad \dots -1 0 1 \dots$	number	
$\quad \text{true} \text{false}$	boolean	
$\quad 'a' \dots$	character	
$\quad \text{"abc"} \dots$	string	
$\text{pat} ::=$	Patterns :	
$\quad \text{string } v$	string	
$\quad \text{character } v$	character	
$\quad \text{boolean } v$	boolean	
$\quad \text{number } v$	number	
$\quad \text{unit } v$	unit	
$\quad \text{any}$	wildcard	
Syntax Metamorphosis		
	$\boxed{\text{term} \rightarrow \text{block}}$	
$b ::=$	Blocks :	
$\quad \text{con} \text{pat} \text{dec} \text{exp}$		
$\quad \text{param} \text{uid} \text{mtc}$		
$\quad \text{field}$		

Typing	$\boxed{\Gamma \vdash b : BT}$
$\boxed{\Gamma \vdash \blacklozenge v : \text{Exp} \otimes T}$ (BT-CONST)	$\boxed{\Gamma \vdash \blacklozenge v : \text{Pat} \otimes T}$ (BT-PAT)

Syntax			Syntax Metamorphosis
$PT ::=$	number string character unit boolean	<i>Primitive types :</i>	$T \rightarrow color$
$T ::=$	$\dots$ PT nothing * $T \rightarrow T$ $PT \Rightarrow T$ (T)	<i>Types:</i> <i>primitive type :</i> <i>nothing</i> <i>any</i> <i>function</i> <i>match – do</i> <i>parentheses</i>	$T ::=$ #778899 #f08080 #808000 #FED049 #228B22 #800000 #008B8B #4363D8 #9A6324
$ST ::=$	Dec Exp Pat Uid Param Field Mtc	<i>Syntax types :</i> <i>declaration</i> <i>expression</i> <i>pattern</i> <i>identifier</i> <i>parameter</i> <i>field</i> <i>match – do</i>	$ST \rightarrow shape$
$BT ::=$	$ST \otimes T$	<i>Block types :</i> <i>ST and T</i>	$ST ::=$
$\Gamma ::=$	$\emptyset$ $\Gamma, b : BT$	<i>Contexts :</i> <i>empty context</i> <i>block binding</i>	<i>Syntax types :</i> <i>declaration</i>
$prog ::=$	MNL <i>dec</i>	<i>Program</i>	
Syntax Metamorphosis			
$prog ::=$		$prog \rightarrow block$	
		<i>Program</i>	

Figure 2: Type

Syntax	Syntax Metamorphosis
$uid ::=$	$\boxed{term \rightarrow block}$
$[ID]^1 letter[letter digit]^*$	$Declaration :$
$param ::=$	$function$
$\mathbf{ID} uid$	$variable$
$ \mathbf{Unit}$	$unit$
$dec ::=$	$Declaration :$
$\mathbf{Function} uid$	$function$
$\mathbf{Parameter} param$	
$\mathbf{Expression} exp$	
$ \mathbf{Variable} uid$	$variable$
$\mathbf{bind\ to} exp$	
$ dec\ dec$	$dec\ sequence$
Syntax Metamorphosis	
$uid ::=$	$\boxed{term \rightarrow block}$
	$Identifier :$
$param ::=$	$Parameter:$
	$unit$
	$variable$
Evaluation	$\boxed{b \rightarrow b'}$
$\frac{b_2 \rightarrow b'_2}{\mathbf{Variable} b_1 \mathbf{bind\ to} b_2 \rightarrow \mathbf{Variable} b_1 \mathbf{bind\ to} b'_2}$	(E-DECBIND)
$\mathbf{Variable} b_1 \mathbf{bind\ to} v_2 \rightarrow [b_1 \mapsto v_2]$	(E-DECV)

Typing

$$\boxed{\Gamma \vdash b : BT}$$

$$\boxed{\Gamma \vdash \mathbf{ID} \ n : Uid \otimes T}$$

(BT-ID)

$$\boxed{\mathbf{Unit} : Param \otimes unit \in \Gamma}$$

(BT-PARAMEMPTY)

$$\boxed{\frac{\Gamma \vdash b_1 : Uid \otimes nothing}{\mathbf{ID} \ b_1 : Param \otimes T \in \Gamma}}$$

(BT-PARAM)

$$\boxed{\frac{\Gamma \vdash b_1 : Uid \otimes nothing \quad \Gamma, b_2 : Param \otimes T_1 \vdash b_3 : Exp \otimes T_2}{\Gamma \vdash \mathbf{Function} \ b_1 \ \mathbf{Parameter} \ b_2. \ \mathbf{Expression} \ b_3 : Dec \otimes T_1 \rightarrow T_2}}$$

(BT-DECFUNC)

$$\boxed{\frac{\Gamma \vdash b_1 : Uid \otimes nothing \quad \Gamma \vdash b_2 : Exp \otimes T}{\Gamma \vdash \mathbf{Variable} \ b_1 \ \mathbf{bind to} \ b_2 : Dec \otimes T}}$$

(BT-DECVAR)

Figure 3: Identifier, Parameter, and Declaration

Syntax		Expressions :	Syntax Metamorphosis
$exp ::=$	$\dots$		$\boxed{term \rightarrow block}$
	<i>con</i>	<i>constant</i>	$exp ::= \dots$ <i>Expressions :</i>
	Lambda	<i>lambda</i>	<i>condition</i>
	Parameter <i>param</i>		
	Expression <i>exp</i>		<i>binary operator</i>
	Application <i>exp</i> ₁	<i>function</i>	
	Over <i>exp</i> ₂	<i>application</i>	
	Bound <i>uid</i>	<i>bound</i>	
	Condition	<i>condition</i>	
	When <i>exp</i>		
	Is true <i>exp</i> ₁		
	Otherwise <i>exp</i> ₂		
	Sequence	<i>sequence</i>	
	Item-1 <i>exp</i> ₁ ;		
	[Item-n <i>exp</i> _n] ⁺	$n \geq 2$	
	Let <i>dec</i>	<i>let binding</i>	
	in	$n \geq 1$	
	[Item-n <i>exp</i> _n] ⁺		
	<i>exp</i> ₁ $\diamond$ <i>exp</i> ₂	<i>binary operator</i>	
	$\diamond$ <i>exp</i> ₁	<i>unary operator</i>	
	Switch <i>exp</i>	<i>case analysis</i>	
	Case-1 <i>mtc</i> ₁ ;		
	[Case-n <i>mtc</i> _n] ⁺	$n \geq 2$	
$mtc ::=$	is <i>pat</i> do <i>exp</i>	<i>Matching :</i>	
Syntax Metamorphosis			
			$\boxed{term \rightarrow block}$
$exp ::=$	$\dots$	<i>Expression</i>	
	<i>con</i>	<i>constant</i>	
		<i>lambda</i>	

Figure 4: Expression - Syntax

Evaluation

$b \rightarrow b'$

$$\frac{b_1 \rightarrow b'_1}{\text{Application of } b_1 \text{ Over } b_2 \rightarrow \text{Application of } b'_1 \text{ Over } b_2} \quad (\text{E-APP1})$$
$$\frac{b_2 \rightarrow b'_2}{\text{Application of } v_1 \text{ Over } b_2 \rightarrow \text{Application of } v_1 \text{ Over } b'_2} \quad (\text{E-APP2})$$
$$\frac{\text{Application of } \lambda x : BT_{11}.b_{12} \text{ Over } v_2}{\rightarrow [x \mapsto v_2]b_{12}} \quad (\text{E-APPABS})$$
$$\text{Condition When } true \text{ Is true } b_2 \text{ Otherwise } b_3 \rightarrow b_2 \quad (\text{E-COND-TRUE})$$
$$\text{Condition When } false \text{ Is true } b_2 \text{ Otherwise } b_3 \rightarrow b_3 \quad (\text{E-COND-FALSE})$$
$$\frac{b_1 \rightarrow b'_1}{\text{Condition When } b_1 \text{ Is true } b_2 \text{ Otherwise } b_3 \rightarrow \text{Condition When } b'_1 \text{ Is true } b_2 \text{ Otherwise } b_3} \quad (\text{E-CONDITION})$$
$$\frac{b_1 \rightarrow b'_1}{\text{Sequence of Exp-1 } b_1 \text{ Exp-2 } b_2 \rightarrow \text{Sequence of Exp-1 } b'_1 \text{ Exp-2 } b_2} \quad (\text{E-SEQ})$$
$$\frac{b_2 \rightarrow b'_2}{\text{Sequence of Exp-1 } unit \text{ Exp-2 } b_2 \rightarrow \text{Sequence of Exp-1 } unit \text{ Exp-2 } b'_2} \quad (\text{E-SEQNEXT})$$
$$\text{Let } b_1 = v_2 \text{ in } b_3 \rightarrow [b_1 \mapsto v_2]b_3 \quad (\text{E-LETV})$$
$$\frac{b \rightarrow b'}{\text{Switch } b \{ \text{Case-i } bc_i^{i \in 1 \dots n} \} \rightarrow \text{Switch } b' \{ \text{Case-i } bc_i^{i \in 1 \dots n} \}} \quad (\text{E-CASEANALYSIS})$$
$$\frac{b \Downarrow v \quad b1_i \Downarrow v1_i \quad v = v1_i}{\text{Switch } b \{ \text{Case-i is } b1_i \text{ do } b2_i^{i \in 1 \dots n} \} \rightarrow b2_i} \quad (\text{E-CASEPROJ})$$
$$\frac{b_2 \rightarrow b'_2}{\text{is } b_1 \text{ do } b_2 \rightarrow \text{is } b_1 \text{ do } b'_2} \quad (\text{E-MATCH})$$

Evaluation

$b \rightarrow b'$

$$\frac{b_1 \rightarrow b'_1}{\text{Terminal } b_1 \text{ operator } b_2 \rightarrow \text{Terminal } b'_1 \text{ operator } b_2} \quad (\text{E-BINOPTR-1})$$
$$\frac{b_2 \rightarrow b'_2}{\text{Terminal } v \text{ operator } b_2 \rightarrow \text{Terminal } v \text{ operator } b'_2} \quad (\text{E-BINOPTR-2})$$
$$\frac{b \rightarrow b'}{\text{Terminal } b \rightarrow \text{Terminal } b'} \quad (\text{E-UNARYOPTR})$$

Typing

$$\boxed{\Gamma \vdash b : BT}$$

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes T_1 \rightarrow T_2 \quad \Gamma \vdash b_2 : Exp \otimes T_1}{\Gamma \vdash \mathbf{Application\ of\ } b_1 \ \mathbf{Over\ } b_2 : Exp \otimes T_2}} \quad (\text{BT-APP})$$

$$\boxed{\frac{\Gamma \vdash b_1 : Uid \otimes nothing \quad b : Param \otimes T \in \Gamma}{\Gamma \vdash \mathbf{Bound\ } b_1 : Exp \otimes T}} \quad (\text{BT-BOUNDPARAM})$$

$$\boxed{\frac{\Gamma \vdash b_1 : Uid \otimes nothing \quad b : Dec \otimes T \in \Gamma}{\Gamma \vdash \mathbf{Bound\ } b_1 : Exp \otimes T}} \quad (\text{BT-BOUNDDEC})$$

$$\boxed{\frac{\Gamma, b_1 : Param \otimes T_1 \vdash b_2 : Exp \otimes T_2}{\Gamma \vdash \mathbf{Lambda\ Parameter\ } b_1 \ \mathbf{Expression\ } b_2 : Exp \otimes T_1 \rightarrow T_2}} \quad (\text{BT-LAMBDA})$$

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes boolean \quad \Gamma \vdash b_2 : Exp \otimes T \quad \Gamma \vdash b_3 : Exp \otimes T}{\Gamma \vdash \mathbf{Condition\ When\ } b_1 \ \mathbf{Is\ true\ } b_2 \ \mathbf{Otherwise\ } b_3 : Exp \otimes T}} \quad (\text{BT-CONDITION})$$

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes unit \quad \Gamma \vdash b_2 : Exp \otimes T_2}{\Gamma \vdash \mathbf{Sequence\ of\ Exp-1\ } b_1 \ \mathbf{Exp-2\ } b_2 : Exp \otimes T_2}} \quad (\text{BT-SEQ})$$

$$\boxed{\frac{\Gamma, b_{dec} \vdash b : Exp \otimes T}{\Gamma \vdash \mathbf{Let\ } b_{dec} \ \mathbf{in\ Exp\ } b : Exp \otimes T}} \quad (\text{BT-LETIN})$$

$$\boxed{\frac{\text{for each } i \quad \Gamma \vdash b : Exp \otimes T_1 \quad \Gamma \vdash bc_i : Mtc \otimes T_1 \implies T_2}{\Gamma \vdash \mathbf{Switch\ } b \{ \mathbf{Case-i\ } bc_i \}_{i \in 1 \dots n} : Exp \otimes T_2}} \quad (\text{BT-CASEANALYSIS})$$

$$\boxed{\frac{\Gamma, b_1 : Pat \otimes T_1 \vdash b_2 : Exp \otimes T_2}{\Gamma \vdash \mathbf{is\ } b_1 \ \mathbf{do\ } b_2 : Mtc \otimes T_1 \implies T_2}} \quad (\text{BT-MATCH})$$

Figure 5: Expression - Evaluation and Typing

Typing

$$\boxed{\Gamma \vdash b : BT}$$

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes T_1 \quad \Gamma \vdash b_2 : Exp \otimes T_1}{\Gamma \vdash \mathbf{Terminal} \ b_1 \ \mathbf{operator} \ b_2 : Exp \otimes T_2}}$$

(BT-BINOPTR)

$$\boxed{\frac{\Gamma \vdash b : Exp \otimes T_1}{\Gamma \vdash \mathbf{Terminal} \ b : Exp \otimes T_2}}$$

(BT-UNARYOPTR)

Figure 6: Expression - Evaluation and Typing

Syntax $ \begin{array}{l} exp ::= \dots \\ \quad \text{ Tuple constructor } \\ \quad \quad [\text{Inhabitant-i } exp_i]^+ \\ \quad \text{ Get Item-}n \\ \quad \quad \text{from Tuple } exp \\ v ::= \dots \\ \quad tuple \{v_i \mid i \in 1..n\} \\ T ::= \dots \\ \quad \text{ tuple of } (T_1[\text{ and } T_n]^*) \end{array} $	Syntax Metamorphosis $T \rightarrow color$ $ \begin{array}{l} T ::= \dots \\ \quad \text{ \#911eb4 } \\ exp ::= \dots \end{array} $ $Types :$ $tuple$ $term \rightarrow block$ $Expressions :$ $constructor$
Evaluation $ \begin{array}{l} tuple \{v_i \mid i \in 1..n\} .j \rightarrow v_j \\ \\ \frac{b \rightarrow b'}{\text{Get Item-}n \text{ from Tuple } b \rightarrow \text{Get Item-}n \text{ from Tuple } b'} \\ \\ \frac{b_j \rightarrow b'_j}{tuple \{v_i \mid i \in 1..j-1, b_j, b_k \mid k \in j+1..n\} \rightarrow tuple \{v_i \mid i \in 1..j-1, b'_j, b_k \mid k \in j+1..n\}} \end{array} $	$b \rightarrow b'$ (E-TUPLEPROJX) (E-TUPLEPROJ) (E-TUPLE)
Typing $\frac{\text{for each } i \quad \Gamma \vdash b_i : Exp \otimes T_i}{\Gamma \vdash \text{ Tuple constructor } \{ \text{Inhabitant-i } b_i \mid i \in 1..n \} : Exp \otimes \text{ tuple of } \{T_i \mid i \in 1..n\}}$ $\frac{\Gamma \vdash b : Exp \otimes \text{ tuple of } \{T_i \mid i \in 1..m\}}{\Gamma \vdash \text{ Get Item- } n \text{ from Tuple } b : Exp \otimes T_n}$	$\Gamma \vdash b : BT$ (BT-TUPLE) (BT-TUPLEPROJ)

Figure 7: Tuple - Syntax and Type

Syntax $ \begin{array}{l} exp ::= \dots \\ \quad \text{Record constructor} \\ \quad \quad [\text{Field-i } field_i]^+ \\ \quad \text{Get field-}n \\ \quad \quad \text{from Record } exp \\ field ::= \\ \quad \text{Field } uid = exp \\ v ::= \dots \\ \quad record \{l_i = v_i \mid i \in 1..n\} \\ T ::= \dots \\ \quad \text{record of} \\ \quad \quad \{l_1 : T_1[\text{and } l_n : T_n]^* \} \end{array} $	Syntax Metamorphosis $\begin{array}{l} T ::= \dots \\ \quad \text{\#FF8C00} \\ exp ::= \dots \end{array}$ $\begin{array}{l} T \rightarrow color \\ Types : \\ \quad record \\ term \rightarrow block \\ Expressions : \\ \quad constructor \end{array}$
Evaluation $ \begin{array}{l} record \{l_i = v_i \mid i \in 1..n\} .l_j \rightarrow v_j \\ \\ \frac{b \rightarrow b'}{\text{Get field-}n \text{ from Record } b} \\ \rightarrow \text{Get field-}n \text{ from Record } b' \\ \\ \frac{b_j \rightarrow b'_j}{record \{l_i = v_i \mid i \in 1..j-1, l_j = b_j, l_k = b_k \mid k \in j+1..n\}} \\ \rightarrow record \{l_i = v_i \mid i \in 1..j-1, l_j = b'_j, l_k = b_k \mid k \in j+1..n\} \end{array} $	$b \rightarrow b'$ (E-RECORDPROJX) (E-RECORDPROJ) (E-RECORD)
Typing $\frac{\Gamma \vdash n : Uid \otimes nothing \quad \Gamma \vdash b : Exp \otimes T}{\Gamma \vdash \text{Field } n = b : Field \otimes n : T}$ $\frac{\text{for each } i \quad \Gamma \vdash b_i : Field \otimes T_i}{\Gamma \vdash \text{Record constructor} \{ \text{Field-i } b_i \mid i \in 1..n \} : Exp \otimes \text{record of } \{T_i \mid i \in 1..n\}}$ $\frac{\Gamma \vdash b : Exp \otimes \text{record of } \{T_i \mid i \in 1..m\}}{\Gamma \vdash \text{Get field- } n \text{ from Record } b : Exp \otimes T_n}$	$\Gamma \vdash b : BT$ (BT-FIELD) (BT-RECORD) (BT-RECORDPROJ)

Figure 8: Record - Syntax and Type

Syntax		Expression :	Syntax Metamorphosis
$exp ::=$	...		$T \rightarrow color$
List constructor			$Types :$
Empty list		<i>empty list</i>	<i>list</i>
List constructor		<i>constructor</i>	$term \rightarrow block$
[Inhabitant-n exp_n]*		$n \geq 1$	$Expressions :$
Is empty exp		<i>test for</i>	<i>empty</i>
		<i>empty list</i>	
Head of List exp		<i>head of a list</i>	
Tail of List exp		<i>tail of a list</i>	
Append exp_1		<i>append a new</i>	
to List exp_2		<i>inhabitant</i>	
$v ::=$	...	$Values :$	
<i>list</i> []		<i>empty list</i>	
<i>list</i> [v_1, v_2]		<i>list constructor</i>	
$T ::=$	...	$Types :$	
list of T		<i>list</i>	
Syntax Metamorphosis			$term \rightarrow block$
$exp ::=$	...	$Expressions :$	<i>constructor</i>

Figure 9: List

Evaluation rules

$b \rightarrow b'$

$$\frac{b_1 \rightarrow b'_1}{\text{Append } b_1 \text{ to List } b_2 \rightarrow \text{Append } b'_1 \text{ to List } b_2} \quad (\text{E-LISTAPPEND1})$$

$$\frac{b_2 \rightarrow b'_2}{\text{Append } b_1 \text{ to List } b_2 \rightarrow \text{Append } b_1 \text{ to List } b'_2} \quad (\text{E-LISTAPPEND2})$$

$$\text{Is empty } list[] \rightarrow true \quad (\text{E-LISTISEMPTYEMPTY})$$

$$\text{Is empty } list[v_1, v_2] \rightarrow false \quad (\text{E-LISTISEMPTYLIST})$$

$$\frac{b_1 \rightarrow b'_1}{\text{Is empty } b_1 \rightarrow \text{Is empty } b'_1} \quad (\text{E-LISTISEMPTY})$$

$$\text{Head of List } list[v_1, v_2] \rightarrow v_1 \quad (\text{E-HEADOFLISTCONS})$$

$$\frac{b_1 \rightarrow b'_1}{\text{Head of List } b_1 \rightarrow \text{Head of List } b'_1} \quad (\text{E-HEADOFLIST})$$

$$\text{Tail of List } list[v_1, v_2] \rightarrow v_2 \quad (\text{E-TAILOFLISTCONS})$$

$$\frac{b_1 \rightarrow b'_1}{\text{Tail of List } b_1 \rightarrow \text{Tail of List } b'_1} \quad (\text{E-TAILOFLIST})$$

Figure 10: List

Typing rules	$\boxed{\Gamma \vdash b : BT}$
$\boxed{\Gamma \vdash \mathbf{List\ constructor\ Empty\ list} : Exp \otimes list\ of\ ^*}$	(BT-LISTEMPTY)
$\frac{\text{for each } i \quad \Gamma \vdash b_i : Exp \otimes T}{\Gamma \vdash \mathbf{List\ constructor} \{ \mathbf{Inhabitant-i}\ } b_i^{i \in 1 \dots n} : Exp \otimes list\ of\ T}$	(BT-LISTCONS)
$\frac{\Gamma \vdash b : Exp \otimes list\ of\ T}{\Gamma \vdash \mathbf{Is\ empty}\ b : Exp \otimes boolean}$	(BT-LISTISEMPTY)
$\frac{\Gamma \vdash b : Exp \otimes list\ of\ T}{\Gamma \vdash \mathbf{Head\ of\ List}\ b : Exp \otimes T}$	(BT-LISTHEAD)
$\frac{\Gamma \vdash b : Exp \otimes list\ of\ T}{\Gamma \vdash \mathbf{Tail\ of\ List}\ b : Exp \otimes list\ of\ T}$	(BT-LISTTAIL)
$\frac{\Gamma \vdash b_1 : Exp \otimes T \quad \Gamma \vdash b_2 : Exp \otimes list\ of\ T}{\Gamma \vdash \mathbf{Append}\ b_1\ \mathbf{to\ List}\ b_2 : Exp \otimes list\ of\ T}$	(BT-LISTAPPEND)

Figure 11: List

Syntax		Syntax Metamorphosis	
$term ::=$	...	$Terms :$	$T \rightarrow color$
	$\lambda X.t$	$type\ abstraction$	
	$t [T]$	$type\ application$	
$v ::=$	...	$Values :$	
	X	$type\ abstraction\ value$	
$T ::=$	...	$Types :$	
	$'X$	$type\ variable$	
	$\forall X.T$	$universal$	
$\Gamma ::=$	...	$Contexts :$	
	$\Gamma, ST \otimes X$	$type\ variable\ binding$	
Evaluation rules		$b \rightarrow b'$	
$(\lambda X.b)[T_2] \rightarrow [X \mapsto T_2]b$		(E-TABS)	
$\frac{b_1 \rightarrow b'_1}{b_1[BT_2] \rightarrow b'_1[BT_2]}$		(E-TAPP)	
Typing rules		$\Gamma \vdash b : BT$	
$\frac{\Gamma, X \vdash b : BT}{\Gamma \vdash \lambda X.b : \forall X.BT}$		(BT-TABS)	
$\frac{\Gamma \vdash b : \forall X.BT_1}{\Gamma \vdash b [ST_2 \otimes T_2] : [X \mapsto T_2]BT_1}$		(BT-TAPP)	

Figure 12: Polymorphic

Constraint typing rules

$$\boxed{\Gamma \vdash b : BT \quad |_X \ C}$$

$$\boxed{\frac{\Gamma \vdash b : \text{Uid} \quad b_1 : ST \otimes T \in \Gamma}{\Gamma \vdash \mathbf{Bound} \ b : \text{Exp} \otimes T \quad |_{\emptyset} \ \{ \}}}$$

(CT-BOUND)

$$\boxed{\Gamma \vdash \blacklozenge v : \text{Exp} \otimes T \quad |_{\emptyset} \ \{ \}}$$

(CT-CONSTANT)

$$\boxed{\Gamma \vdash \blacklozenge v : \text{Pat} \otimes T \quad |_{\emptyset} \ \{ \}}$$

(CT-PATTERN)

$$\boxed{\frac{\begin{array}{c} \Gamma \vdash b_1 : \text{Uid} \otimes \text{nothing} \\ \Gamma, b_2 : T_2 \vdash b_3 : \text{Exp} \otimes T_3 \quad |_X \ C \end{array}}{\Gamma \vdash \mathbf{Function} \ b_1 \ \mathbf{Parameter} \ b_2 \\ \mathbf{Expression} \ b_3 : \text{Exp} \otimes T_2 \rightarrow T_3 \quad |_X \ C}}$$

(CT-DECFUNC)

$$\boxed{\frac{\Gamma, b_1 : T_1 \vdash b_2 : \text{Exp} \otimes T_2 \quad |_X \ C}{\Gamma \vdash \mathbf{Lambda} \ \mathbf{Parameter} \ b_1 \ \mathbf{Expression} \ b_2 : \text{Exp} \otimes T_1 \rightarrow T_2 \quad |_X \ C}}$$

(CT-LAMBDA)

$$\boxed{\frac{\begin{array}{c} \Gamma \vdash b_1 : \text{Exp} \otimes T_1 \quad |_{X_1} \ C_1 \quad \Gamma \vdash b_2 : \text{Exp} \otimes T_2 \quad |_{X_2} \ C_2 \\ X_1 \cap X_2 = X_1 \cap FV(T_2) = X_2 \cap FV(T_1) = \emptyset \\ T \notin X_1, X_2, T_1, T_2, C_1, C_2, \Gamma, b_1, \text{orb}_2 \\ C' = C_1 \cup C_2 \cup \{T_1 = T_2 \rightarrow T\} \end{array}}{\Gamma \vdash \mathbf{Application} \ b_1 \ \mathbf{Over} \ b_2 : \text{Exp} \otimes T \quad |_{X_1 \cup X_2 \cup X} \ C}}$$

(CT-APP)

$$\boxed{\frac{\begin{array}{c} \Gamma \vdash b_1 : \text{Exp} \otimes T_1 \quad |_{X_1} \ C_1 \quad \Gamma \vdash b_2 : \text{Exp} \otimes T_2 \quad |_{X_2} \ C_2 \\ \Gamma \vdash b_3 : \text{Exp} \otimes T_3 \quad |_{X_3} \ C_3 \\ X_1, X_2, X_3 \text{ non overlapping} \\ C' = C_1 \cup C_2 \cup C_3 \cup \{T_1 = \text{boolean}, T_2 = T_3\} \end{array}}{\Gamma \vdash \mathbf{Condition} \ \mathbf{When} \ b_1 \ \mathbf{Is true} \ b_2 \ \mathbf{Otherwise} \ b_3 \\ : \text{Exp} \otimes T_2 \quad |_{X_1 \cup X_2 \cup X_3} \ C'}}$$

(CT-COND)

$$\boxed{\frac{\begin{array}{c} \Gamma \vdash b_1 : \text{Exp} \otimes T_1 \quad |_{X_1} \ C_1 \quad \Gamma \vdash b_2 : \text{Exp} \otimes T_2 \quad |_{X_2} \ C_2 \\ X_1, X_2 \text{ non overlapping} \quad C' = C_1 \cup C_2 \end{array}}{\Gamma \vdash \mathbf{Sequence of Exp-1} \ b_1 \ \mathbf{Exp-2} \ b_2 : \text{Exp} \otimes T_2 \quad |_{X_1 \cup X_2} \ C_1}}$$

(CT-SEQ)

Figure 13: Constraint typing rule

Constraint typing rules

$$\boxed{\Gamma \vdash b : BT \quad |_X C}$$

(CT-PRIMITIVEBINARYOPERATOR)

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes T_1 \quad |_{X_1} C_1 \quad \Gamma \vdash b_2 : Exp \otimes T_2 \quad |_{X_2} C_2 \quad C' = C_1 \cup C_2 \cup \{T_1 = T_2 = \text{Primitive type}\}}{\Gamma \vdash \mathbf{Terminal} \ b_1 \ \mathbf{operator} \ b_2 : Exp \otimes T \quad |_{X_1 \cup X_2} C'}}$$

(CT-PRIMITIVEUNARYOPERATOR)

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes T_1 \quad |_{X_1} C_1 \quad C' = C_1 \cup \{T_1 = \text{Primitive type}\}}{\Gamma \vdash \mathbf{Terminal} \ b_1 : Exp \otimes T \quad |_{X_1} C'}}$$

$$\boxed{\frac{\Gamma \vdash [b_1 \mapsto b_2]b_3 : Exp \otimes T_3 \quad |_{X_1} C_1 \quad \Gamma \vdash b_2 : Exp \otimes T_2 \quad |_{X_2} C_2 \quad X_1, X_2 \text{ non overlapping} \quad C' = C_1 \cup C_2}{\Gamma \vdash \mathbf{Let} \ b_1 = b_2 \ \mathbf{in} \ b_3 : Exp \otimes T_3 \quad |_{X_1 \cup X_2} C'}}$$

(CT-LETPOLY)

$$\boxed{\frac{\begin{array}{l} \text{for each } i \quad \Gamma \vdash b : Exp \otimes T_1 \quad |_{X_1} C_1 \\ \Gamma \vdash bc_i : Mtc \otimes T_{1i} \implies T_{2i} \quad |_{X_{2i}} C_{2i} \\ X_1, X_{2i} \text{ non overlapping} \\ C' = \bigcup C_{2i} \cup C_1 \cup \{T_1 = T_{1i}, T_2 = T_{2i}\} \end{array}}{\Gamma \vdash \mathbf{Switch} \ b \ \{\mathbf{Case-i} \ bc_i \ ^{i \in 1 \dots n}\} : Exp \otimes T_2 \quad |_{\bigcup X_i} C'}}$$

(CT-CASEANALYSIS)

$$\boxed{\frac{\begin{array}{l} \text{for each } i \quad \Gamma \vdash b_i : Exp \otimes T_i \quad |_{X_i} C_i \\ C' = \bigcup C_i \cup \{T = T_i\} \end{array}}{\Gamma \vdash \mathbf{List constructor} \ \{\mathbf{Inhabitant-i} \ b_i \ ^{i \in 1 \dots n}\} : Exp \otimes \text{list of } T \quad |_{\bigcup X_i} C'}}$$

(CT-LISTCONS)

$$\boxed{\frac{\Gamma \vdash b_1 : Exp \otimes T_1 \quad |_{X_1} C_1 \quad \Gamma \vdash b_2 : Exp \otimes \text{list of } T_2 \quad |_{X_2} C_2 \quad X_1, X_2 \text{ non overlapping} \quad C' = C_1 \cup C_2 \cup \{T_1 = T_2\}}{\Gamma \vdash \mathbf{Append} \ b_1 \ \mathbf{to List} \ b_2 : Exp \otimes \text{list of } T_1 \quad |_{X_1 \cup X_2} C'}}$$

(CT-LISTAPPEND)

$$\boxed{\frac{\Gamma \vdash b : Exp \otimes \text{list of } T_1 \quad |_{X_1} C_1 \quad C' = C_1 \cup \{T_1 = T\}}{\Gamma \vdash \mathbf{Head of List} \ b : Exp \otimes T \quad |_{X_1}^{18} C'}}$$

(CT-LISTHEAD)

Figure 14: Constraint typing rule

Constraint typing rules

$$\boxed{\Gamma \vdash b : BT \quad |_X C}$$

$$\frac{\Gamma \vdash b : Exp \otimes list\ of\ T_1 \quad |_{X_1} C_1 \quad C' = C_1 \cup \{T_1 = T\}}{\Gamma \vdash \mathbf{Tail\ of\ List}\ b : Exp \otimes list\ of\ T \quad |_{X_1} C'}$$

(CT-LISTTAIL)

$$\frac{\Gamma \vdash b : Exp \otimes T \quad |_X C \quad C' = C \cup \{T = list\ of\ X\}}{\Gamma \vdash \mathbf{Is\ empty}\ b : Exp \otimes boolean \quad |_X C'}$$

(CT-LISTISEMPTY)

$$\frac{\text{for each } i \quad \Gamma \vdash b_i : Exp \otimes T_i \quad |_{X_i} C_i \quad X_i \text{ non overlapping} \quad C' = \bigcup C_i}{\Gamma \vdash \mathbf{Tuple\ constructor} \quad \{\mathbf{Inhabitant-i}\ b_i^{i \in 1 \dots n}\} : Exp \otimes \mathbf{tuple\ of}\ \{T_i^{i \in 1 \dots n}\} \quad |_{\bigcup X_i} C'}$$

(CT-TUPLE)

$$\frac{\text{for each } i \quad \Gamma \vdash b : Exp \otimes \mathbf{tuple\ of}\ \{T_i^{i \in 1 \dots m}\} \quad |_{X_i} C_i \quad C' = C_{i=n} \cup \{T_{i=n} = T\}}{\Gamma \vdash \mathbf{Get\ Item-}\ n \mathbf{\ from\ Tuple}\ b : Exp \otimes T \quad |_{X_{i=n}} C'}$$

(CT-TUPLEPROJ)

$$\frac{\text{for each } i \quad \Gamma \vdash b_i : Field \otimes T_i \quad |_{X_i} C_i \quad X_i \text{ non overlapping} \quad C' = \bigcup C_i}{\Gamma \vdash \mathbf{Record\ constructor} \quad \{\mathbf{Field-i}\ b_i^{i \in 1 \dots n}\} : Exp \otimes \mathbf{record\ of}\ \{T_i^{i \in 1 \dots n}\} \quad |_{\bigcup X_i} C'}$$

(CT-RECORD)

$$\frac{\Gamma \vdash b : Exp \otimes \mathbf{record\ of}\ \{T_i^{i \in 1 \dots m}\} \quad |_{X_i} C_i \quad C' = C_{i=n} \cup \{T_{i=n} = T\}}{\Gamma \vdash \mathbf{Get\ field-}\ n \mathbf{\ from\ Record}\ b : Exp \otimes T \quad |_{X_{i=n}} C'}$$

(CT-RECORDPROJ)

Figure 15: Constraint typing rule